

PRODUCT REQUIREMENT DOCUMENT (PRD)

UMHackathon 2026

Group Name: Last minute

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Table of Contents

1. Project Overview
2. Background & Business Objectives
3. Product Purpose
4. System Functionalities
5. User Stories & Use Cases
6. Features Included (Scope Definition)
7. Features Not Included (Scope Control)
8. Assumptions & Constraints
9. Risks & Questions Throughout Development

Project Name: StitchFlow

Version: 2

1. Project Overview

- **Problem Statement:** Event and operations teams process highly fragmented, sensitive, and unstructured inputs such as contracts, emails, forms, chats, screenshots, PDFs, and spreadsheet notes. Manual handling is slow and error-prone, while fully autonomous AI processing introduces serious risks around hallucination, privacy, compliance, and incorrect execution. Teams therefore need a secure and auditable middle layer that can parse complex inputs, screen for policy issues, reconcile conflicting evidence, and prepare structured execution drafts without removing the human from the final decision.
- **Target Domain:** AI Systems & Enterprise Workflow Automation, with emphasis on: Secure document and message understanding, Human-in-the-loop orchestration, Policy and compliance screening, Structured workflow execution support, Multi-source operational coordination.
- **Proposed Solution Summary:** StitchFlow V2 is an AI-powered, human-in-the-loop decision-support engine. Using Z.AI's GLM-4.5 as the central reasoning engine, StitchFlow ingests messy inputs, applies redaction before reasoning, screens for policy risks, compares documents, detects ambiguity and contradictions, and outputs structured, reviewable actions such as Google Sheets rows, dashboard items, clause annotations, and Gmail drafts for human approval.

2. Background & Business Objective

- **Background of the Problem:** In modern enterprise environments—particularly within event management and large-scale operations—data does not arrive in neat, machine-readable formats. Instead, it exists as a fragmented mosaic of unstructured inputs: contractual PDFs, rapid-fire chat logs, scanned screenshots, and disparate email threads. For example, teams spend thousands of labor hours manually reconciling

conflicting data points (e.g., a date in an email vs. a date in a contract). This is not only slow and unscalable but introduces a high "fatigue tax," where human error leads to costly operational failures.

- **Importance of Solving This Issue:** In domains like legal and compliance, the cost of a "hallucinated" clause or a missed policy violation can reach millions in regulatory fines. StitchFlow V2 acts as a secure audit layer, ensuring that AI never makes a definitive legal or financial judgment in a vacuum. It provides the evidence, but leaves the authority with the human expert.
- **Strategic Fit / Impact:** AI workflow automation can be both powerful and enterprise-safe. It leverages GLM-4.5's massive 128K to 131K context window to hold policy references, precedent T&Cs, extracted document content, and user instructions together during workflow reasoning.

3. Product Purpose

3.1. Main Goal of the System: To provide a secure, AI-enabled workflow coordinator that transforms messy multi-source operational inputs into structured, reviewable decisions, while screening for compliance risks, surfacing ambiguity, and preserving human control over execution.

3.2. Intended Users (Target Audience):

- Event Directors & Ops Managers
- Legal & Compliance Reviewers
- Administrative Coordinators
- Project and service operations teams
- Student committees and SME ops staff

4. System Functionalities

4.1. Description: The system operates as a Stateful Human-in-the-Loop (HITL) Engine, flowing through the following architecture: Ingest → Redact → Reason (GLM) → Validate → Structure → Human Review → Execute → Log. It receives unstructured inputs from users or documents, sends them to GLM for reasoning, converts them into structured workflow metadata, and then applies workflow rules to decide what actions should be executed next. The product is designed to support ambiguity handling, missing information detection, escalation, and structured output generation rather than only simple classification or storage.

4.2. Key Functionalities:

- **Vibe Designing:** The system has the ability to translate human-language descriptive instructions directly into high-fidelity designs.
- **Conversational Editing:** It facilitates the continuous modification of an existing design by allowing the user to speak directly to the canvas itself or by texting instructions.
- **Versioning & Infinite Canvas:** The platform features an AI Agent-enabled version tracking system that provides dozens of design variation generations within the workspace.
- **Export- Component Based:** The system enables the seamless conversion of visual designs into functional code snippets, as well as into editable Figma tool layers.

4.3. AI Model & Prompt Design

4.3.1. Model Selection: The selected model is Z.AI GLM (e.g. GLM-4.5 or later) because it supports deep natural language understanding, multi-turn reasoning, instruction parsing, and conversion of human language into executable tasks. These capabilities make it suitable for workflow scenarios where the system must interpret messy input, reason across multiple variables, and produce structured outputs for downstream execution.

4.3.2. Prompting Strategy: The team uses a multi-step agentic prompting strategy with structured output constraints. The system asks GLM to classify the input, extract entities, detect ambiguity, recommend the next action, and return the result in a strict JSON schema.

4.3.3. Context & Input Handling: Long inputs are handled by truncating low-value text, chunking into smaller sections, or rejecting the file with guidance if it is too large for safe processing.

4.3.4. Fallback & Failure Behavior: If the model returns off-topic or malformed output, the system validates the JSON schema, retries once with a stricter prompt, and sends the item to human review if confidence stays low.

5. User Stories & Use Cases

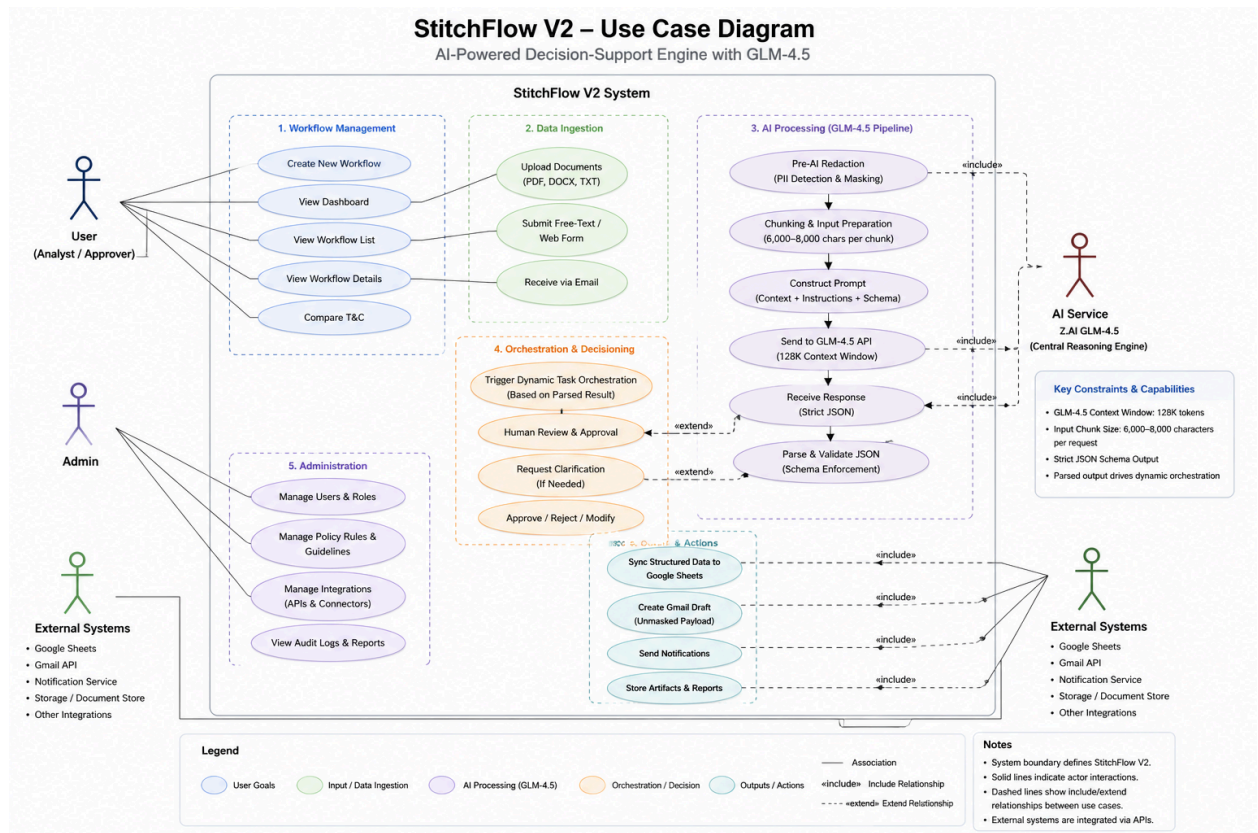
- **User Stories:**

- a. As a Compliance Officer, I want uploaded documents to be screened against our policy library with quoted evidence and confidence, so I can review possible issues quickly.
- b. As an Ops Manager, I want AI extractions to populate Google Sheets as pending-review rows, so I can approve or reject actions systematically.
- c. As an Event Planner, I want to upload multiple past event T&Cs and receive a new draft with precedent clauses, AI recommendations, and legal-review sections clearly distinguished.
- d. As a Coordinator, I want the AI to prepare Gmail drafts for vendor or stakeholder communication, so I retain full control over what is actually sent.
- e. As a reviewer, I want contradictions and missing fields to surface before execution, so incomplete or conflicting requests do not move forward silently.

- **Use Cases (Main Interactions):**

- Input-to-Workflow Flow:** User submits content, GLM interprets it, extracts entities, detects issues, and recommends the next action.
- Clarification Flow:** Missing information causes the system to generate a clarification request and hold the item in the proper state.
- Operational Dashboard Flow:** Users review all workflow items by state, confidence, urgency, and review status.
- Failure Recovery Flow:** If model output or an API step fails, the item moves into retry or escalation rather than being silently dropped.

- **Use Case Diagram:**



6. Features Included (Scope Definition)

- Ingestion & Redaction: File upload with automatic masking of sensitive data before reasoning.
- Z.AI GLM-4.5 Reasoning Layer: Multi-step prompt chain for extraction, contradiction checks, policy screening, and next-action recommendation.
- Policy Screening Dashboard: UI showing flagged issues, confidence scores, and source evidence.
- Google Sheets Sync: AI-generated rows for review and state management.
- Gmail API Integration: Draft creation for outbound communication developers.
- T&C Comparison UI: Color-coded text interface for precedent, generated, and legal-review clauses.
- Clarification Flow: Missing-data and ambiguity handling.
- Stateful Workflow Engine: Explicit item states and transitions.
- Retry & Escalation Logic: Failure recovery for model or integration errors.
- Audit Trail: Logging of decisions, human overrides, and workflow transitions.

7. Features Not Included (Scope Control)

- Fully autonomous messaging or execution
- Automatic legal or regulatory certification
- Autonomous approval of high-risk policy cases
- Post-processing-only redaction
- Full production-grade WhatsApp automation
- Large-scale model fine-tuning
- Long-term predictive analytics

8. Assumptions & Constraints

- **LLM Cost Constraint:** To manage inference cost, the system caches repeated system context or repeated content where possible. External references show GLM-5.1 pricing at about \$0.60 per million input tokens, making this feasible.
- **Context Constraints:** The large context window supports processing precedent files, policy references, and workflow instructions in a single reasoning pass, within practical bounds.

- **Prototype Integration Limits:** Connectors may be constrained to demo-level configurations, and some data inputs may rely on extracted text rather than full production OCR pipelines.
- **User Input:** Execution speed is intentionally limited by human review for sensitive or high-impact steps.

9. Risks & Questions Throughout Development

- **Risk:** False Positives in Policy Screening, where the AI may flag lawful activities due to ambiguous wording.
- **Risk:** Privacy Exposure.
- **Risk:** Silent Failure in API Orchestration.